

Figure 1: Program flow

The above program calls `func()` the simple way. Let's modify the program to call using a pointer to a function. Here's the changed `main()` function:

```
#include<stdio.h>
int func(int, int);
int main(void)
{
    int result1,result2;
    /* declaring a pointer to a function which takes
       two int arguments and returns an integer as result */
    int (*ptrFunc)(int,int);

    /* assigning ptrFunc to func's address */
    ptrFunc=func;

    /* calling func() through explicit dereference */
    result1 = (*ptrFunc)(10,20);

    /* calling func() through implicit dereference */
    result2 = ptrFunc(10,20);
    printf("result1 = %d result2 = %d\n",result1,result2);
    return 0;
}

int func(int x, int y)
{
    return x*y;
}
```

The output has no surprises:

```
result1 = 30 result2 = 30
```

A simple callback function

At this stage, we have enough knowledge to deal with function callbacks. According to Wikipedia, "In computer programming, a callback is a reference to executable code, or a piece of executable code, that is passed as an argument to other code. This allows a lower-level software layer to call a subroutine (or function) defined in a higher-level layer." Let's try one simple program to demonstrate this. The complete program has three

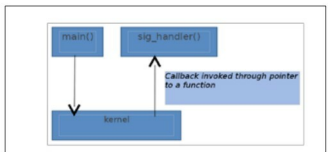


Figure 2: Kernel callback

files: `callback.c`, `reg_callback.h` and `reg_callback.c`.

```
/* callback.c */
#include<stdio.h>
#include"reg_callback.h"

/* callback function definition goes here */
void my_callback(void)
{
    printf("inside my_callback\n");
}

int main(void)
{
    /* initialize function pointer to
       my_callback */
    callback ptr_my_callback=my_callback;
    printf("This is a program demonstrating function callback\n");
    /* register our callback function */
    register_callback(ptr_my_callback);
    printf("back inside main program\n");
    return 0;
}

/* reg_callback.h */
typedef void (*callback)(void);
void register_callback(callback ptr_reg_callback);

/* reg_callback.c */
#include<stdio.h>
#include"reg_callback.h"

/* registration goes here */
void register_callback(callback ptr_reg_callback)
{
    printf("inside register_callback\n");
    /* calling our callback function my_callback */
    (*ptr_reg_callback());
}
```